

General vs. Individual NPDES Permits: Regulatory Framework and Empirical Patterns in ICIS-NPDES

July 21, 2026

Scope

This brief distinguishes the Clean Water Act’s two NPDES permitting vehicles – **individual** and **general** permits – on regulatory grounds, then shows how that distinction is encoded in the ICIS-NPDES (ECHO) bulk data used in this project, and how it shapes observed enforcement patterns. All data figures are computed by `make_permit_types_brief.R` from the raw ICIS-NPDES files and the existing enforcement breakdown in `output/enforcement_by_permit_type.csv` (`scripts/enforcement_by_permit_type.R`); see §5.

1 The regulatory distinction

Individual permits. An individual permit is written for one specific facility. Its effluent limits combine technology-based limits (set nationally, by industry category) with **water-quality-based limits derived from the specific receiving water**, so two dischargers with identical processes can face different numeric limits depending on where they discharge. Individual permits require a full, facility-specific application (40 CFR § 122.21) and carry facility-specific monitoring and Discharge Monitoring Report (DMR) obligations. They are the norm for larger, more complex dischargers – municipal treatment plants and major industrial sources.

General permits. A general permit (40 CFR § 122.28) is a single template permit the permitting authority issues once to cover **many similar, lower-risk dischargers** at once – construction and industrial stormwater, concentrated animal feeding operations (CAFOs), and small or short-lived operations. A facility obtains coverage by filing a **Notice of Intent (NOI)** against the master permit, which substitutes for an individual application; conditions are standardized and monitoring is lighter or absent. The relationship is one-to-many: one master general permit can cover hundreds of thousands of individual coverages (§2).

A common misconception: permit term. Both permit types are capped at the *same* statutory maximum: 40 CFR § 122.46 limits any NPDES permit, individual or general, to a fixed term not exceeding five years. The practical difference is administrative, not legal – reissuing one general permit covers every facility under it at once, so general permits are almost always issued for the full five years, while an individual permit’s specific term (and whether it is reissued on time or instead runs on under administrative continuance) depends on that facility’s own permitting history (see `docs/data_quirks.md` on the ADC status flag).

Moving between vehicles. The Director may require any discharger covered by a general permit to apply for and obtain an individual permit instead (40 CFR § 122.28(b)(3)) – e.g., if a facility’s risk profile turns out not to fit the general permit’s assumptions. Conversely, an owner/operator

covered by a general permit may itself request exclusion and apply for an individual permit within 90 days of the general permit’s publication.

Representative general-permit programs. EPA’s Multi-Sector General Permit (MSGP) covers stormwater discharges from industrial facilities across roughly thirty sectors in areas where EPA is the permitting authority; the Construction General Permit (CGP) covers stormwater from construction sites disturbing ≥ 1 acre with a pathway to a water of the US. CAFOs are typically covered under state- or EPA-issued general permits as well. These are the concrete real-world instances of the “construction and industrial stormwater, CAFOs” category.

2 How the distinction appears in ICIS-NPDES

PERMIT_TYPE_CODE marks the permitting vehicle on every ICIS-NPDES permit record. NPD = individual permit; GPC = a facility’s coverage under a general permit; NGP = the general permit itself (the master record, not a covered facility) – a small correction to an earlier internal note that had mislabeled this as GPG, a code that does not appear anywhere in the raw file. The remaining codes (UFT, IIU, APR, SIN, SNN) are minor, low-volume variants (state-issued or non-NPDES-equivalent records); their exact definitions were not independently confirmed against an authoritative EPA reference table and are not load-bearing for this analysis.

Table 1: PERMIT_TYPE_CODE frequency, two framings: **Permits** = distinct EXTERNAL_PERMIT_NMBR values (one row per permit); **Records** = raw rows in ICIS_PERMITS.csv (one row per permit *version* – reissuances and modifications each add a row).

Code	Permits (distinct)	Records (all versions)
GPC	1,029,662	1,368,680
NPD	99,939	242,678
UFT	46,360	46,360
IIU	7,315	11,105
APR	5,901	16,382
SIN	3,573	6,562
NGP	1,232	2,823
SNN	41	56

The gap between the two columns matters: general-permit *coverages* (GPC) outnumber individual permits by roughly ten to one on either count, but reissuance behavior differs – 51.9% of individual permits carry more than one version (a permit keeps its NPDES_ID across ~ 5 -year reissuances), versus 21.8% of general-permit coverages, which are more often **renumbered** when the master permit is reissued (e.g., an NDR100000-cycle coverage becomes an NDR110000-cycle coverage under a new ID) rather than versioned under the same one.

Major status tracks the individual vehicle. Of 8,062 permits ever flagged “major” in any version, 7,800 (96.8%) are individual permits. Restricting analysis to major *and* individual permits, as this project’s facility panel does, is therefore a modest restriction on the major population and a severe one on the permit universe as a whole (99,939 individual permits of 1,194,023 total).

Facility identity and shared IDs. Individual permits map cleanly to one physical site with a stable FRS registry identifier (FACILITY_UIN) and point geocode. General-permit coverages frequently **share one generic** FACILITY_UIN across unrelated sites: the most extreme case in this data is one North Dakota UIN (110070099629) spanning 361 distinct general-permit coverages (construction-stormwater projects at different locations, sharing an administrative facility record rather than a physical one). This makes facility-level aggregation reliable for individual permits and hazardous for general ones.

Both at once. 10,362 FACILITY_UINs hold both an individual (NPD) and a general (GPC) permit. This is typically an individually permitted process discharge *plus* a general stormwater coverage at the same site – the two permits govern different discharges, not the same one, so they should not be collapsed into a single “one facility, one permit” record.

Data richness. Monitoring and enforcement are concentrated on the individual/major side. Per docs/data_quirks.md, FY2025 DMR data shows majors at ~84% reporting coverage versus ~7% for minors – consistent with general permits’ lighter-to-absent numeric monitoring requirements.

3 Enforcement patterns by permit type

Table 2: Formal vs. informal enforcement action records, by permit type and major/minor status (all years, all actions in the bulk enforcement files; each row is one action record, not one facility).

Permit type	Status	Formal	Informal	Total	% Informal
Individual	Minor	42,307	351,145	393,452	89.2%
General	Minor	23,313	237,934	261,247	91.1%
Individual	Major	30,942	179,192	210,134	85.3%
Other	Unknown	12,748	27,885	40,633	68.6%
Other	Minor	1,406	24,147	25,553	94.5%
General	Unknown	513	427	940	45.4%
Other	Major	279	526	805	65.3%
General	Major	59	596	655	91.0%
Individual	Unknown	249	125	374	33.4%

Pooling across major/minor status, individual permits accrue 603,960 action records (12.2% formal / 87.8% informal) and general permits 262,842 (9.1% formal / 90.9% informal). The two permit types look broadly similar in their formal/informal *split*, but arrive there by different detection pathways:

- **Individual permits** (mostly majors) generate violations chiefly through routine DMR review (numeric effluent-limit exceedances) and scheduled inspections. Individual/major permittees are also the population to which EPA’s **significant noncompliance (SNC)** criteria are principally applied: since 1995, EPA policy has defined SNC to include, among other triggers, any monthly average effluent limit exceeded by $\geq 40\%$ (or by $\geq 20\%$ for two or more consecutive months), any limit violated for four or more months in a two-consecutive-quarter window, unauthorized bypass/pass-through, and reporting violations 30+ days late. These criteria presuppose numeric monthly DMR limits – exactly what individual permits carry and general permits typically do not – which is why SNC tracking is substantively an individual/major-permit concept rather than a general-permit one.

- **General permits** have less routine numeric monitoring, so violations more often surface through inspections, failure to file the NOI or other required paperwork, or complaints, rather than a DMR exceedance. Consistent with this, the response is more often informal (90.9% vs. 87.8% for individual). The Director’s authority to revoke general-permit coverage and require an individual permit (§ 1 above; 40 CFR § 122.28(b)(3)) is the formal escalation path when a covered facility turns out to warrant closer, facility-specific regulation.

4 The 2015 NPDES Electronic Reporting Rule

EPA’s NPDES Electronic Reporting Rule took effect **December 21, 2015**, replacing paper-based NPDES reporting with mandatory electronic submission, with the stated goal of more complete, timely, and nationally consistent compliance data. Implementation was phased: **Phase 1** (largely individual-permit DMR data) had a compliance deadline of December 21, 2016; **Phase 2** (general-permit and other remaining data streams) was originally due December 21, 2020 but was later extended by rule, giving affected programs up to five additional years. This phased rollout is the mechanism behind the “2016 eRule” reporting break already flagged in this project’s data documentation (`docs/data_quirks.md`): majors’ DMR coverage was already high before the rule, while minors’ and general-permit coverage only began rising as Phase 1, and later Phase 2, requirements took hold – so pre- and post-2016 (and pre/post-2020+) coverage are not directly comparable, particularly for general-permit-covered dischargers.

5 Implications for analysis

For facility-level research, the individual permit is the meaningful analytical unit: one physical site, a stable identifier, dense measurement, and a trustworthy location. The general permit is best understood as an **administrative umbrella, not a facility** – a single FACILITY_UIN or master permit number can span hundreds of unrelated sites, and monitoring/enforcement data is comparatively sparse. Restricting to major individual permits, as this project’s panel does, buys a well-defined and well-measured population at a real cost to external validity: it excludes the large majority of permitted dischargers, which are general, lightly monitored, and (per § 3) usually resolved informally rather than through the DMR/inspection/SNC pipeline that individual permits generate.

Reproducibility

Every figure above is generated by `make_permit_types_brief.R` from `data/raw/npdes_downloads/ICIS_PERMITS.csv`, `data/raw/npdes_downloads/ICIS_FACILITIES.csv`, and the pre-built `output/enforcement_by_permit_type.csv` (from `scripts/enforcement_by_permit_type.R`, not recomputed here). Regulatory citations (40 CFR §§ 122.28, 122.46; the 1995 SNC criteria memorandum; the 2015 Electronic Reporting Rule and its Phase 2 extension) are drawn from the Federal Register and eCFR and are not derived from project data. Rerun the generator after any raw-data refresh; do not hand-edit the `\input` fragments or this PDF.